

The Renewal Threshold

Predicting PG Resident Renewal Intention
Using Behavioral and Service-Experience Analytics

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1. Problem Statement and Objectives

1.1 Problem Statement

The Paying Guest (PG) accommodation and co-living sector operates in a highly competitive environment where resident retention is an important contributor to long-term business sustainability. When residents leave after a relatively short stay, operators may incur repeated acquisition, onboarding, administrative, and vacancy-related costs. Consequently, understanding the factors associated with a resident's intention to renew is important for improving service quality and reducing avoidable churn.

PG operators traditionally focus on physical characteristics such as room quality, internet connectivity, air conditioning, furnishings, security, and other amenities to attract and retain residents. However, the resident experience extends beyond physical infrastructure. Operational factors such as the number of complaints raised, speed of complaint resolution, deposit refund transparency, and perceived fairness of rent can influence how residents evaluate the overall value of their accommodation.

This case study therefore investigates whether service-experience variables provide greater predictive information about renewal intention than physical amenity and security variables. The analysis compares three feature groups: an Amenity-only model, a Service-only model, and a Combined model.

1.2 Business Significance

Resident retention has direct operational importance for PG management. If service-related experiences are strongly associated with renewal intention, management can use complaint and service information to identify residents who may require intervention.

A complaint management system could identify residents who repeatedly report maintenance problems. Instead of waiting until the end of the rental period, management could intervene earlier by improving resolution speed, communicating progress, or escalating unresolved issues. Similarly, understanding rent fairness and deposit transparency can help management improve the perceived value of accommodation.

1.3 Objectives

- Descriptive and Exploratory Analysis: To examine the characteristics of the resident survey data and identify patterns between resident experience variables and renewal intention.
- Predictive Modeling and Feature Evaluation: To compare the predictive performance of Amenity-only, Service-only, and Combined feature blocks using Logistic Regression and Random Forest models.
- Business Recommendation: To translate analytical findings into practical recommendations related to complaint management, service quality, rent fairness, and resident retention.

1.4 Research Questions

- Does service experience provide more predictive information about renewal intention than amenity and security ratings?
- Which resident experience variables are most strongly associated with renewal intention?
- Does combining service and amenity variables improve predictive performance?
- How can PG operators use these findings to design proactive resident-retention strategies?

2. Data Collection and Dataset Description

2.1 Data Collection

The analytical study uses a structured survey-style dataset containing 10,000 records. The dataset represents responses that could be collected through a questionnaire from PG residents. The questionnaire structure captures demographic, financial, accommodation, service-experience, satisfaction, and renewal-intention information. The analysis is focused on understanding patterns in the supplied survey records rather than making claims about the entire PG accommodation population.

2.2 Dataset Characteristics

Category	Variable	Description
Demographic	Occupation	Student or working professional
Tenure	Stay Duration	Duration of residence in months
Financial	Monthly Rent	Current monthly rental amount
Physical Experience	Amenities Rating	Rating from 1–5
Physical Experience	Security Rating	Rating from 1–5
Service Experience	Complaint Count	Number of complaints/issues raised
Service Experience	Complaint Resolution	Speed/status of complaint resolution
Financial Experience	Deposit Refundability	Perception of deposit refund process
Financial Experience	Rent Fairness	Perception of whether rent is fair
Satisfaction	Overall Satisfaction	Overall resident satisfaction score
Target	Renewal Intention	Yes, Maybe, or No
Qualitative	Biggest Reason	Main reason stated by the resident

2.3 Renewal Intention Distribution

Renewal Intention	Records	Percentage
Yes	4,706	47.06%
Maybe	4,194	41.94%
No	1,100	11.00%
Total	10,000	100%

The largest individual category is Yes, representing 47.06% of the records. However, the large proportion of Maybe responses is also important because it indicates uncertainty among a substantial portion of residents.

For binary classification, Maybe responses are excluded. This produces a modeling dataset containing 5,806 definite Yes/No responses. Within this subset, Yes represents approximately 81.05% and No approximately 18.95%, creating a class-imbalance problem. Class-balanced machine-learning models are therefore used rather than relying on ordinary accuracy.

3. Data Preparation and Exploratory Analysis

3.1 Data Cleaning

The dataset was examined for missing values, duplicates, data types, and inconsistent categorical representations. The analytical preparation included checking missing values, checking duplicate records, converting numerical variables to appropriate formats, standardizing categorical text values, encoding ordinal categorical variables, separating the three renewal categories for descriptive analysis, creating a binary Yes/No target for predictive modeling, and excluding variables that could introduce target leakage.

3.2 Encoding of Variables

Rent Fairness: Underpriced = 0, Fair = 1, Overpriced = 2. **Deposit Refundability:** Straightforward = 0, Not sure = 1, Disputes common = 2. Complaint resolution was represented according to resolution speed: Same day = 0, Few days = 1, Over a week = 2, Never resolved = 3, with No complaints raised treated as a separate model category.

3.3 Target Leakage Prevention

Biggest Reason is a free-text variable and was excluded from predictive modeling because it may directly contain the respondent's explanation for their renewal decision. Overall Satisfaction was also excluded from the primary predictive feature comparison because it can behave as a proxy for renewal intention and may artificially inflate

predictive performance.

3.4 Exploratory Analysis

Amenity Rating

A positive relationship was observed between amenity ratings and renewal intention. At an amenities rating of 2, approximately 32.4% of residents expressed definite Yes renewal intention. At a rating of 5, this increased to approximately 54.9%. This indicates that physical amenities are relevant to renewal intention, but amenities alone do not explain the complete retention pattern.

Complaint Count

Complaint Count	Definite Yes Renewal
0	64.3%
2	32.9%
4	10.6%

Complaint volume shows a pronounced relationship with renewal intention. The observed renewal rate falls substantially as complaint volume increases. From a business perspective, complaint count can therefore be treated as a potential early-warning indicator. The relationship should be interpreted as an association in the analytical data rather than a causal effect.

Rent Fairness

Residents who considered their rent Fair had a definite renewal rate of approximately 49.5%, whereas residents who considered the rent Overpriced had a rate of approximately 20.2%. This indicates that residents evaluate accommodation not only according to absolute rent but also according to their perception of the value received.

Correlation Analysis

A Spearman correlation analysis was used for numerical and ordinal variables. The analysis is primarily descriptive because ordinal codes represent ordering rather than equal numerical distances. Correlation does not establish causation.

4. Analytics Method and Implementation

4.1 Analytical Framework

The analytical process follows a structured Business Analytics workflow: Data → Preparation → Exploratory Analysis → Statistical Analysis → Feature-Block Modeling → Model Evaluation → Feature Importance → Business Recommendations.

4.2 Mann–Whitney U Test

The Mann–Whitney U test was used to compare numerical or ordinal feature distributions between definite Yes and No groups. Variables examined include amenities, security, complaint count, complaint resolution, deposit concern, rent unfairness, and overall satisfaction. A rank-biserial effect size was also considered to provide an indication of the magnitude and direction of group differences.

4.3 Chi-Square Test

For categorical variables such as occupation, a Chi-square test of association was used. Cramér's V was also considered as an effect-size measure. Because the dataset is synthetic, statistical significance should be interpreted cautiously.

4.4 Logistic Regression

Logistic Regression was used as an interpretable baseline predictive model. The model estimates the probability of a binary outcome and allows the direction and magnitude of coefficients to be examined.

4.5 Random Forest

Random Forest was selected as a non-linear ensemble model. It can capture complex relationships and interactions between variables and therefore complements the interpretability of Logistic Regression.

4.6 Cross-Validation and Class Imbalance

The study uses Repeated Stratified 5-Fold Cross-Validation with 10 repetitions, producing 50 held-out validation folds. Class-balanced learning strategies were used because the definite Yes/No dataset contains approximately 81.05% Yes and 18.95% No.

The primary evaluation metrics are ROC-AUC, Average Precision, Balanced Accuracy, and F1-score. Ordinary accuracy is not used as the sole performance measure because a majority-class model can achieve approximately 81% accuracy while failing to identify the minority class.

5. Comparison with State-of-the-Art Methods

Published Study / Year	Dataset	Method Used	Evaluation Metric	Key Result	Comparison with Your Work
In (2024)	Commercial real-estate tenant context	Influence/hypothesis testing	Statistical significance	Management reliability, adequacy and tenant empathy associated with perceived value and renewal intention.	Similar renewal problem; this study converts service dimensions into measurable predictive features and evaluates them with cross-validation.
Marandi, Tasavori & Najmi (2024)	Approx. 100,000 TripAdvisor reviews from hotels	Machine learning, text analysis and customer segmentation	Text-feature and segment analysis	71 features identified; room, staff, food and accessibility were salient.	Uses large real-world unstructured review data; this study uses structured PG survey-style variables.
Park & Lee (2025)	Survey of single-person households in Busan	Hypothesis testing and mediation analysis	Significance and mediation analysis	Location factors associated with lease renewal; satisfaction mediated selected relationships.	Both examine residential renewal this study focuses on PG service and operational variables.

The studies use different populations, variables, outcomes, and methodologies. Their numerical results should therefore not be directly ranked against the present model results. The comparison instead focuses on methodological differences, data characteristics, findings, strengths, and limitations.

6. Results, Business Insights and Recommendations

6.1 Model Performance

Feature Block	Logistic Regression ROC-AUC	Random Forest ROC-AUC
Amenity Only	0.660	0.654
Service Only	0.787	0.806
Combined	0.831	0.842

The Service-only feature block produces higher ROC-AUC values than the Amenity-only block for both algorithms. The Combined feature block produces the highest predictive performance among the tested configurations. Random Forest achieves a slightly higher ROC-AUC than Logistic Regression for the Service-only and Combined feature sets.

6.2 Baseline, Balanced Accuracy and Average Precision

The majority-class baseline produces approximately 0.811 accuracy but a Balanced Accuracy of 0.500. The Combined Random Forest achieved a Balanced Accuracy of approximately 0.762 and an Average Precision of approximately 0.950. These results indicate useful discriminative and ranking performance within the analytical data, while not establishing equivalent performance on real-world PG residents.

6.3 Feature Importance

Permutation importance identified Complaint Count as the most prominent feature in the combined Random Forest analysis, with an observed ROC-AUC decrease of approximately 0.223 when permuted. Amenities and rent

unfairness followed with approximate decreases of 0.048 and 0.037 respectively. This importance analysis is descriptive and should not be interpreted as a causal ranking.

6.4 Business Insights

- **Complaint volume as an early-warning signal:** Repeated complaints are associated with substantially lower definite renewal intention and can be incorporated into retention monitoring.
- **Resolution speed matters:** Complaint count should be considered together with how quickly and effectively issues are resolved.
- **Perceived value matters:** The difference between Fair and Overpriced rent perceptions indicates the importance of transparent value communication.
- **Amenities remain relevant:** Physical infrastructure contributes to resident experience, but the model comparison indicates that service variables contain more predictive information in this analysis.

6.5 Recommendations

Recommendation 1: Complaint-Threshold Alert System

Introduce automated alerts for residents who repeatedly report service problems. A possible operational framework is one complaint for the standard workflow, two complaints for enhanced monitoring, and three or more for management escalation. The exact threshold should be validated using future real-world data.

Recommendation 2: Service-Level Agreements

Establish measurable service-resolution targets covering first response time, resolution time, repeat complaint rate, unresolved complaints, resident complaint rate, and escalation rate.

Recommendation 3: Rent and Deposit Transparency

Provide residents with clear information about rent, included utilities, internet, maintenance, security, deposit amount, refund conditions, and additional charges. This can reduce uncertainty and improve perceived value.

Recommendation 4: Resident Experience Dashboard

Combine renewal intention, complaint count, satisfaction, rent fairness, resolution time, open complaints, amenities, and security into a management dashboard.

Recommendation 5: Prospective Validation

Future analysis should validate these relationships using newly collected, anonymized resident data and actual renewal outcomes where ethically and operationally appropriate.

7. Conclusion and References

7.1 Conclusion

This case study examined factors associated with PG resident renewal intention using behavioral, service-experience, financial-perception, and physical-property variables. The analysis combined exploratory analysis, statistical association testing, Logistic Regression, and Random Forest modeling.

The predictive comparison showed that Service-only features achieved higher ROC-AUC than Amenity-only features for both algorithms. The Combined Random Forest achieved the highest reported ROC-AUC of 0.842, with Balanced Accuracy of approximately 0.762 and Average Precision of approximately 0.950.

Complaint Count emerged as the most prominent variable in the permutation-importance analysis. Exploratory analysis also showed substantial differences in renewal intention across complaint levels and perceived rent fairness categories.

Overall, the results suggest that service-experience variables contain substantial predictive information about renewal intention in the analyzed data, while amenities remain a relevant component of the resident experience. From a

business perspective, complaint monitoring, faster resolution, transparent pricing and deposit policies, and resident-experience dashboards provide practical areas for management attention.

The findings should nevertheless be interpreted as exploratory. Model performance on the supplied analytical data does not establish causal relationships or guarantee equivalent performance for real PG residents. Future work should validate the findings using independently collected real-world observations and actual renewal outcomes.

7.2 Limitations

- The analysis uses a synthetic survey dataset rather than a verified population sample of real PG residents.
- The binary model excludes Maybe responses, removing an important category of resident uncertainty.
- The analysis is cross-sectional and cannot establish causality.
- Ordinal categories are encoded for analysis and should not be interpreted as equally spaced numerical measurements.
- Feature importance can be affected by relationships among correlated variables.
- Model performance on generated data may reflect the underlying data-generation structure.
- Actual renewal behavior was not available as the primary outcome; the analysis focuses on stated renewal intention.
- Future validation on independently collected real-world observations is necessary.

7.3 References

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